

CPSC-354 Programming Languages (Fall 2026)

Quiz 1 — Natural Number Game (NNG) (*Mock*)

Name: _____ Section: _____ Date: _____

Time: 30 minutes

Total points: 10

Instructions. Answer all questions in the spaces provided. Show all reasoning. No calculators, phones, or notes except one letter-sized handwritten reference sheet prepared by the student.

Name	Definition
one_eq_succ_zero	$1 = S(0)$
two_eq_succ_one	$2 = S(1)$
three_eq_succ_two	$3 = S(2)$
four_eq_succ_three	$4 = S(3)$
add_zero	$a + 0 = a$
add_succ	$a + S(b) = S(a + b)$
succ_add	$S(a) + b = S(a + b)$

Figure 1: Names and Definitions

1. Lean to mathematics (5 points).

The following Lean proof establishes that $2 + 2 = 4$ in Tutorial World Level 8. Translate it into an ordinary mathematical proof. Do not use induction; this proof only rewrites definitions. Justify each step using the names from Figure 1.

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example : 2 + 2 = 4 :=
  rw [four_eq_succ_three]
  rw [three_eq_succ_two]
  rw [two_eq_succ_one]
  rw [one_eq_succ_zero]
  rw [add_succ]
  rw [add_succ]
  rw [add_zero]
  rfl
```

2. Proof by induction (5 points).

Prove by induction on b that for all natural numbers a , b , and c ,

$$(a + b) + c = (a + c) + b.$$

Justify each step using the names from Figure 1.